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We have discussed in the previous chapter how food provides energy for growth and repair of tissues. As mentioned earlier along with food, oxygen is necessary for breakdown of glucose to energy. In this chapter we shall discuss the respiratory organs of human, the mechanism of breathing, exchange and transport of gases and a few respiratory disorders.

The term respiration refers to the exchange of oxygen and carbondioxide between environment and cells of our body where organic nutrients are broken down enzymatically to release energy.

6.1 Respiratory functions

The five primary functions of the respiratory system are –

i. To exchange O_2 and CO_2 between the atmosphere and the blood.
ii. To maintain homeostatic regulation of body pH.
iii. To protect us from inhaled pathogens and pollutants.
iv. To maintain the vocal cords for normal communication (vocalization).
v. To remove the heat produced during cellular respiration through breathing.

6.2 Respiratory organs in various organisms.

Different animals have different organs for exchange of gases, depending upon their habitats and levels of organization. The amount of dissolved oxygen is very low in water compared to the amount of oxygen in the air. So the rate of breathing in aquatic organisms is much faster than land animals.

In animals like sponges, coelenterates and flatworms exchange of gases takes place through the body surface by simple diffusion. Earthworms use their moist skin, whereas insects have tracheal tubes. Gills are used as respiratory organs in most of the aquatic Arthropods and Molluscs. Among verterbrates, fishes use gills whereas amphibians, reptiles, birds and mammals have well vascularised lungs. Frogs spend most of their time in water and also use their moist skin for respiration along with lungs.

6.2.1 Human Respiratory System

The respiratory system includes the external nostrils, nasal cavity, the pharynx, the larynx, the trachea, the bronchi and bronchioles and the lungs which contain the alveoli (Figure 6.1). The parts starting from the external nostrils up to the terminal bronchioles constitute the conducting zone, whereas the alveoli and the ducts are called the respiratory zone. The parts of the conducting zone, humidifies and warms the incoming air.

In human beings, air enters the upper respiratory tract through the external nostrils. The air passing through the nostrils is filtered by fine hairs and mucus lining the passage. The external nostrils lead to the nasal chamber which opens into the nasopharynx which opens through the glottis of the larynx region into the trachea. The ciliated epithelial cells lining the trachea, bronchi and bronchioles secrete mucus. Mucus membrane lining the airway contains goblet cells which secrete mucus, a slimy material rich in glycoprotein. Microorganisms and dust particles attach in the mucus films and